

Assignment: Linear Models

Q4.

$$\textcircled{1} \quad SSE = \sum_{i=1}^N e_i^2 = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

We are given $\hat{y}_i = b_0 + b_1 z_{i1} + b_2 z_{i2}$

$$SSE = \sum_{i=1}^N (y_i - (b_0 + b_1 z_{i1} + b_2 z_{i2}))^2$$

$$SSE = \sum_{i=1}^N (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2})^2$$

$$\textcircled{2} \quad \frac{\partial SSE}{\partial b_0} = \sum_{i=1}^N 2(y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) \cdot -1$$

$$\frac{\partial SSE}{\partial b_0} = -2 \sum_{i=1}^N (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2})$$

$$\frac{\partial SSE}{\partial b_1} = \sum_{i=1}^N 2(y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) \cdot (-z_{i1})$$

$$\frac{\partial SSE}{\partial b_1} = -2 \sum_{i=1}^N z_{i1} (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2})$$

$$\frac{\partial SSE}{\partial b_2} = \sum_{i=1}^N 2(y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) \cdot (-z_{i2})$$

$$\frac{\partial SSE}{\partial b_2} = -2 \sum_{i=1}^N z_{i2} (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2})$$

- ③ At the optimum, the partial derivatives from ② are equal to 0.

$$\frac{\partial SSE}{\partial b_0} = -2 \sum_{i=1}^N (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) = 0$$

$$\sum_{i=1}^N (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) = 0$$

$$\sum_{i=1}^N (y_i - \hat{y}_i) = \sum_{i=1}^N e_i = 0$$

so, $\boxed{\frac{1}{N} \sum_{i=1}^N e_i = 0}$ and the average error is 0 at the optimum.

$$\frac{\partial SSE}{\partial b_1} = -2 \sum_{i=1}^N z_{i1} (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) = 0$$

$$\sum_{i=1}^N z_{i1} (y_i - \hat{y}_i) = 0$$

$$\sum_{i=1}^N z_{i1} e_i = z_1 e = e z_1 = 0$$

$$\frac{\partial SSE}{\partial b_2} = -2 \sum_{i=1}^N z_{i2} (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) = 0$$

$$\sum_{i=1}^N z_{i2} (y_i - \hat{y}_i) = 0$$

$$\sum_{i=1}^N z_{i2} e_i = z_2 e = e z_2 = 0$$

so, $\boxed{e \cdot z = 0 \text{ at the optimum.}}$

④ Taking the average error is 0 at the optimum:

$$\frac{1}{N} \sum_{i=1}^N e_i = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i) = 0$$

$$\frac{1}{N} \sum_{i=1}^N (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) = 0$$

$$\frac{1}{N} \sum_{i=1}^N (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) = 0$$

$$\frac{1}{N} \sum_{i=1}^N y_i - b_0 - \frac{b_1}{N} \sum_{i=1}^N z_{i1} - \frac{b_2}{N} \sum_{i=1}^N z_{i2} = 0$$

We assume z_{ij} is centered, so $\frac{1}{N} \sum_{i=1}^N z_{i1} = 0$ and $\frac{1}{N} \sum_{i=1}^N z_{i2} = 0$.

$$\frac{1}{N} \sum_{i=1}^N y_i - b_0 - \cancel{\frac{b_1}{N} \sum_{i=1}^N z_{i1}}^0 - \cancel{\frac{b_2}{N} \sum_{i=1}^N z_{i2}}^0 = 0$$

$$\frac{1}{N} \sum_{i=1}^N y_i - b_0 = 0$$

$$b_0 = \frac{1}{N} \sum_{i=1}^N y_i = \bar{y}$$

The optimal intercept is $b_0^* = \bar{y}$.

Eliminating b_0 from the other equations:

$$\sum_{i=1}^N z_{i1} (y_i - \bar{y} - b_1 z_{i1} - b_2 z_{i2}) = 0$$

$$\sum_{i=1}^N z_{i2} (y_i - \bar{y} - b_1 z_{i1} - b_2 z_{i2}) = 0$$

$$\sum_{i=1}^N z_{i1} (y_i - \bar{y} - b_1 z_{i1} - b_2 z_{i2}) = 0$$

$$\sum_{i=1}^N z_{i1} (y_i - \bar{y}) - \sum_{i=1}^N b_1 z_{i1}^2 - \sum_{i=1}^N b_2 z_{i1} z_{i2} = 0$$

$$\sum_{i=1}^N z_{i1} (y_i - \bar{y}) = \sum_{i=1}^N b_1 z_{i1}^2 + \sum_{i=1}^N b_2 z_{i1} z_{i2}$$

$$\sum_{i=1}^N z_{i1} (y_i - \bar{y}) = b_1 \sum_{i=1}^N z_{i1}^2 + b_2 \sum_{i=1}^N z_{i1} z_{i2}$$

$$\sum_{i=1}^N z_{i2} (y_i - \bar{y} - b_1 z_{i1} - b_2 z_{i2}) = 0$$

$$\sum_{i=1}^N z_{i2} (y_i - \bar{y}) - \sum_{i=1}^N b_1 z_{i1} z_{i2} - \sum_{i=1}^N b_2 z_{i2}^2 = 0$$

$$\sum_{i=1}^N z_{i2} (y_i - \bar{y}) = \sum_{i=1}^N b_1 z_{i1} z_{i2} + \sum_{i=1}^N b_2 z_{i2}^2$$

$$\sum_{i=1}^N z_{i2} (y_i - \bar{y}) = b_1 \sum_{i=1}^N z_{i1} z_{i2} + b_2 \sum_{i=1}^N z_{i2}^2$$

$$\textcircled{5} \quad A\mathbf{b} = \mathbf{C}$$

$$\underbrace{\begin{pmatrix} \sum_{i=1}^N z_{i1}^2 & \sum_{i=1}^N z_{i1} z_{i2} \\ \sum_{i=1}^N z_{i1} z_{i2} & \sum_{i=1}^N z_{i2}^2 \end{pmatrix}}_A \underbrace{\begin{pmatrix} b_1 \\ b_2 \end{pmatrix}}_{\mathbf{b}} = \underbrace{\begin{pmatrix} \sum_{i=1}^N z_{i1} (y_i - \bar{y}) \\ \sum_{i=1}^N z_{i2} (y_i - \bar{y}) \end{pmatrix}}_{\mathbf{C}}$$

⑥

$$\frac{1}{N} \begin{pmatrix} \sum_{i=1}^N z_{i1}^2 & \sum_{i=1}^N z_{i1} z_{i2} \\ \sum_{i=1}^N z_{i1} z_{i2} & \sum_{i=1}^N z_{i2}^2 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \frac{1}{N} \begin{pmatrix} \sum_{i=1}^N z_{i1} (y_i - \bar{y}) \\ \sum_{i=1}^N z_{i2} (y_i - \bar{y}) \end{pmatrix}$$

$$z_{i1} = x_{i1} - m_1 \quad z_{i2} = x_{i2} - m_2$$

$$A = \begin{pmatrix} \frac{1}{N} \sum_{i=1}^N (x_{i1} - \bar{x}_1)^2 & \frac{1}{N} \sum_{i=1}^N (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2) \\ \frac{1}{N} \sum_{i=1}^N (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2) & \frac{1}{N} \sum_{i=1}^N (x_{i2} - \bar{x}_2)^2 \end{pmatrix}$$

$$C = \begin{pmatrix} \frac{1}{N} \sum_{i=1}^N (x_{i1} - \bar{x}_1)(y_i - \bar{y}) \\ \frac{1}{N} \sum_{i=1}^N (x_{i2} - \bar{x}_2)(y_i - \bar{y}) \end{pmatrix}$$

We can see that these are variances and covariances.

$$A = \begin{pmatrix} \text{VAR}(x_1) & \text{COV}(x_1, x_2) \\ \text{COV}(x_1, x_2) & \text{VAR}(x_2) \end{pmatrix}$$

$$C = \begin{pmatrix} \text{COV}(x_1, y) \\ \text{COV}(x_2, y) \end{pmatrix}$$

Matrix A is intuitively how much variance and the direction of linear relationships between the explanatory variables.

Matrix C is intuitively how each explanatory variable moves with the response variable.